

Stage 8

Paper 1 Non-fiction

Insert



40 years and counting: the team behind Voyager's space odyssey

On a chilly March morning, Steve Howard, aged 65, is at work in his office in Altadena, California. Two computer screens are squeezed on to his corner desk along with family photos, a box of paper tissues, and tins of peppermints. The office is in a quiet business park. Next to it is a diner, where people linger for hours over a \$1 coffee. If the few people walking by on West Woodbury Road were to peer into Howard's office, they might guess, seeing the graph-covered twin screens and a third PC at the other end of the desk, that he was, perhaps, a financial adviser. But what Steve Howard is actually doing makes this very ordinary scene quite extraordinary. 5

Howard is a NASA¹ mission controller. He is sending instructions to a probe in interstellar space, billions of kilometres from Earth. The 815-kilogram craft, Voyager 1, is one of two identical machines that for many years now have been the furthest human-made objects from Earth. 10

It is no hyperbole to say, then, that the man tapping away at his keyboard is a key figure in the greatest-ever feat of human exploration. There was nothing like the Voyager 1 and Voyager 2 missions to the outer planets before they launched in 1977. The machines have been travelling at around 60 000 kilometres per hour for over 40 years. The Voyagers' on-board computers are early 1970s models that were advanced then but are puny now – a smartphone's computer is some 200 000 times faster and has about 250 000 times more memory than Voyager's hardware. 15 20

The Voyager mission, originally meant to last four years, took the craft initially to Jupiter, then Saturn, then, since everything was working well, to Uranus and finally Neptune, after which they spun off into their journey around the Milky Way. 'We all knew we were on a mission of discovery,' says Professor Ed Stone, aged 79. 'We just had no idea how much discovery there would be. We just kept finding things we didn't know were there to be found. For example, before Voyager, the only known volcanoes in the solar system were here on Earth. Then we flew by Jupiter's moon, Io, which had 10 times the volcanic activity of Earth. Ten times! We detected hot lakes of lava on the surface. That was the first major discovery and it set the tone for the rest of the mission.' 25

Against all expectations their vintage electronics are still, mostly, working in the intense minus 253 °C cold of outer space, but the on-board camera on each Voyager was deactivated to save power in 1990. This was after Voyager 1 took a now-iconic 'family portrait' of the solar system from almost 6.5 billion kilometres out. It captured Neptune, Uranus, Saturn, Jupiter, Venus, Earth (described by the astrophysicist Carl Sagan as a 'pale blue dot') and the Sun, by then just a tiny point of light. 30 35

Engineers are not given to emotion, but the romance of this incredible voyage of discovery has, by their own account, kept the ageing mission team together. Even latecomers, who were at school when Voyager was launched, have been working on the same mission for 30 years and more.

It is clear talking to Voyager staff that they genuinely love their spacecraft, even though most were too young to see them before they flew. But as engineers, they have mixed feelings about the most famous aspect of that romance, the 'golden record' that each craft carries. This is a gold-covered copper disk containing, in groove form, 115 photos from Earth, a selection of natural sounds from surf to whales, music from a variety of cultures and eras and spoken greetings in 55 languages. 40 45

Carl Sagan, who had the initial idea for the record, wrote in the 1970s: 'The spacecraft will be encountered and the recording played only if there are advanced spacefaring civilisations in

interstellar space.' Sagan's son Nick, then an infant, now a science-fiction novelist and screenwriter, recorded the English message: 'Hello from the children of planet Earth.' But one sure to bring people to tears is the message in Mandarin: 'Hope everyone's well. We are thinking about you all. Please come here to visit when you have time.'

50

Project manager Suzy Dodd's view of the record is more typical of the team. 'I'm of the opinion that space is very empty and the chances of something finding it are remote. But that doesn't diminish the fact that we've got a little time capsule out there travelling through space and now orbiting around in our galaxy. And that's us.'

55

For the most part, Voyager is the reality of space exploration – slow, patient science, humdrum perhaps, but real. It's only a 20-minute drive from Altadena to Hollywood, where brilliant fake versions of space exploration are confected, but Voyager, starring real people who keep tissues and tins of peppermints on their desks, is surely one of the most amazing things in human history.

60

Glossary:

¹NASA: the US organisation that is responsible for space exploration

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English

Stage 8

Paper 1 Non-fiction

2024

Cambridge Lower Secondary Progression Test

Name

Class

Date _____

1 hour 10 minutes

Additional materials: Insert

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should pay attention to punctuation, spelling and handwriting.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].
- Suggestions for how long to spend on each section are given in the booklet.

Section A: Reading

Spend 35 minutes on this section.

Read the **text**, written by a journalist after he visited a NASA mission controller, in the insert, and answer Questions 1–10.

1 Look at the first paragraph (lines 2–9).

- (a) Steve Howard does not have a lot of room on his table. Give **one** word that tells the reader this.

..... [1]

- (b) Steve's office is in an unexciting location. Explain **two** ways that the writer shows this. Support each way with a quotation.

The writer does this by:

Quotation 1:

The writer does this by:

Quotation 2:

[4]

- (c) Look at the last sentence. What does the writer use this sentence to do?
Tick (✓) **one** box.

to add more detail to the description of Steve's office

☐

to emphasise a contrast between the first and second paragraphs

☐

to draw a conclusion based on the evidence in the first paragraph

☐

to introduce other colleagues working in Steve's office in the second paragraph

☐

[1]

2 Look at the second paragraph (lines 10–13).

- (a) The writer begins the paragraph with a short sentence about Steve's job. What effect does this have?

..... [1]

- (b) The second sentence refers back to an opinion in the first paragraph. What is the opinion from the first paragraph?

..... [1]

3 Look at the third paragraph (lines 14–20).

- (a) The writer believes Voyager is *the greatest-ever feat of human exploration*. Give **one** phrase that tells the reader this.

..... [1]

- (b) What does the word *puny* tell the reader?

..... [1]

4 Look at the fourth paragraph (lines 21–29).

- (a) How does the writer build up detail in the first sentence?

..... [1]

- (b) An exclamation mark (!) is used to show how the speaker feels. What feeling is this? Tick (✓) **one** box.

joy ☐

horror ☐

disgust ☐

wonder ☐

[1]

- 5 Look at the fifth paragraph (lines 30–35).
Inverted commas (‘ ’) are used twice in two different ways. What are they?
Tick (✓) **two** boxes.

to show words that disagree with the writer’s point of view

☐

to show words borrowed from another language

☐

to show words that usually mean something different

☐

to show words spoken or written by someone

☐

to show words that have been made up

☐

[2]

- 6 Look at the sixth and seventh paragraphs (lines 36–45).
Which idea links these paragraphs?
Tick (✓) **one** box.

Although the team members joined the mission at different times,

they were all at school together.

☐

they agree on everything except one part of it.

☐

they have worked together on other missions for over 30 years.

☐

they all witnessed the launch of the mission.

☐

[1]

- 7 Look at the eighth paragraph (lines 46–51).
Who are the messages on the recording intended for?

..... [1]

- 8 Look at the ninth paragraph (lines 52–55).
Who does the speaker mean when she says *us*?

..... [1]

9 Look at the tenth paragraph (lines 56–60).

(a) Explain the contrast between two places that the writer makes in this paragraph.

.....
..... [2]

(b) Give **one** phrase from the final paragraph that links it back to the first paragraph.

..... [1]

10 Look at lines 10–45.

(a) Complete this table with information from the text.

Voyager space mission information	
Date of launch	
Purpose of original mission	
Equipment carried on board spacecraft	
Achievements of the mission	- discovery: - image: - journey:
Current position	

[3]

(b) Using the information from your table, write a summary of the Voyager mission. Write up to **50 words**. Use your own words as much as possible.

.....

.....

.....

.....

..... [2]

Section B: Writing

Spend 35 minutes on this section.

- 11** Write a report for your school's online magazine about some teamwork that you took part in.

You should consider:

- what you and your team had to do, and why
- what you were able to achieve as a team, and what you might do differently in future
- how the experience affected you and other team members.

Space for your plan:

Write your report on the next page.

[25 marks]

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Stage 8

Cambridge Lower Secondary Progression Test

Insert



Text for Section A, an extract from *Neverwhere* by Neil Gaiman

While trying to find his way home, a young man called Richard encounters the Marquis¹ de Carabas.

De Carabas grinned to himself like a hungry panther sighting a lost peasant child. He knelt down, and took a small metal object from a pocket, which he pushed into a manhole cover at the edge of the alley and twisted. The manhole cover came up, easily; the Marquis put away the metal object, and took something out of another pocket that reminded Richard a little of a long firework or a flare. He held it in one hand, ran his other hand along it, and the far end erupted into scarlet flame.

5

‘Can I ask a question?’ said Richard.

10

‘Certainly not,’ said the Marquis. ‘You don’t ask any questions. You don’t get any answers. You don’t stray from the path. You don’t even think about what’s happening to you right now. Got it?’

‘But—?’

‘Most important of all: no buts. Now, time is of the essence. Move.’ He pointed into the depths revealed by the open manhole cover. Richard moved, clambering down the metal ladder set into the wall beneath the manhole.

15

Richard wondered where they were. He walked nervously, worried that he’d stumble in the darkness and break his ankle. De Carabas strode on ahead, nonchalantly, apparently uncaring of whether Richard was with him or not. The crimson flame cast huge shadows on the tunnel walls.

20

De Carabas put the flare down on the ground, where it continued to sputter and flame, and he began to climb up some metal rungs set into the wall. Richard hesitated, and then followed him. The scarlet light from below was flickering, and then it went out. They climbed in total darkness.

‘So, are we going back?’ he asked.

‘Eventually. And when we get into daylight, don’t look down.’

25

It was daylight (how was it daylight? a voice asked, in the back of his head. It had been almost night when he entered the alley, what, an hour ago?), and he was holding on to a metal ladder that ran up the outside of a very high building (but a few seconds ago he was climbing up the same ladder, and he had been inside, hadn’t he?), and below him he could see...

Tiny cars. Tiny buses and taxis. Tiny buildings. Trees. Miniature lorries. Tiny, tiny people. They swam in and out of focus beneath him.

30

Richard froze on the ladder. His hands clamped tightly to the rungs. His eyes hurt, somewhere behind the eyeballs. He started breathing too fast, too deeply. ‘Somebody,’ said an amused voice above him, ‘wasn’t listening, was he?’

‘I...’ Richard’s throat didn’t work. He swallowed, moistening it. ‘I can’t move.’

35

‘Of course you can move. Or, if you don’t you can stay here, hanging on to the side of the wall.’ Richard looked up at the Marquis. He was looking down at Richard, and still smiling; when he saw that Richard was watching him he let go of the rungs with both his hands, and waggled his fingers at him.

Richard let go of the rung with his right hand and moved it up eight inches², until it found the next rung. Then he moved his right leg up one rung. Then he did it again, with his left hand. After a while he found himself at the edge of a flat roof, and he stepped over it, and collapsed. 40

Richard pulled himself up into a sitting position. They were on the roof of an old building, built of brown stone, with a tower above them. From far below he could hear the wail of a police siren, and the muted roar of traffic. 45

The Marquis prodded Richard gently with his square-toed black boot. ‘Right,’ he said. ‘We better get a move-on, hadn’t we?’ He strode off across the roof and Richard got to his feet and followed, keeping well away from the side of the building. The Marquis opened a door in the side of the tower, and they went down a poorly lit spiral staircase.

It was now completely dark, and Richard stumbled as he reached the last of the steps and found himself looking for a step that wasn’t there. ‘Mind your head,’ said the Marquis, and he opened a door. Richard banged his forehead into something hard, and said ‘ow’, and then stepped out, through a low door, shielding his eyes against the light. 50

Richard rubbed his forehead, then he rubbed his eyes. The door they had just come through was the door to the broom cupboard in the stairwell of his apartment building. It was filled with brooms, and dusters, and an elderly mop, and a huge variety of cleaning fluids, powders and waxes. It had no stairs at the back of it, that he could see, just a wall, on which a stained old calendar hung, quite uselessly, unless 1979 ever came back round. 55

Glossary

¹Marquis: a nobleman

²eight inches: about 20 centimetres

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English

Stage 8

Paper 2 Fiction

2024

Cambridge Lower Secondary Progression Test

Name

Class

Date

1 hour 10 minutes

Additional materials: Insert

INSTRUCTIONS

- Answer **all** questions.
- Write your answer to each question in the space provided.
- You should pay attention to punctuation, spelling and handwriting.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].
- Suggestions for how long to spend on each section are given in the booklet.

Section A: Reading

Spend 35 minutes on this section.

Read the **text** in the insert, and answer Questions 1–7.

1 Look at lines 4–9.

(a) The first sentence uses a simile.

What does the simile tell the reader about the Marquis?

Tick (✓) **one** box.

He has a sense of humour.

☐

He could be dangerous.

☐

He is fond of animals.

☐

He sounds greedy.

☐

[1]

(b) Explain the function and effect of the semi-colon (;) in the first paragraph.

Function:

Effect:

[2]

(c) What does the word *erupted* tell the reader?

..... [1]

2 Look at lines 10–16.

- (a) Explain **two** ways that the Marquis's way of speaking shows that he is unfriendly towards Richard. Support each explanation with a quotation.

First explanation:

Quotation:

Second explanation:

Quotation:

[4]

- (b) Why does the writer repeat the verb *move*?

..... [1]

3 Look at lines 17–25.

- (a) Give **two** words that show the contrast in behaviour between the Marquis and Richard.

The Marquis:

Richard:

[2]

- (b) Give **two** words that show that the flare is gradually weakening.

•

•

[2]

4 Look at lines 26–42.

- (a) The word *tiny* is repeated to emphasise how small things appear to Richard. What else does the repetition of *tiny* emphasise?

..... [1]

- (b) Which statement below do you agree with most?

Tick (✓) **one** box.

This part of the text shows how frightened Richard is.

☐

This part of the text shows how unkind the Marquis is.

☐

Give **two** reasons for your choice. Support each reason with a quotation.

Reason 1:

Quotation 1:

Reason 2:

Quotation 2:

[4]

5 Look at lines 43–53.

Richard has difficulty keeping up with the Marquis. Give **two** phrases that tell the reader this.

•

.....

•

.....

[2]

6 Look at lines 54–58. Why does Richard rub his eyes?

Tick (✓) **one** box.

He is uncertain where he is.

☐

He has been dazzled by the light.

☐

He is astonished by what he sees.

☐

He has just bumped his head.

☐

[1]

- 7 Look at the whole text.
Some of the things that happen in the text are unusual. Give **two** reasons why you think Richard could be surprised. Support each reason with a quotation.
Complete the table below. One example has been done for you.

Reason for surprise	Quotation
<i>The Marquis knows how to get into the tunnel</i>	<i>The manhole cover came up easily...</i>
1	
2	

[4]

Section B: Writing

Spend 35 minutes on this section.

- 8** Write a story about two characters who are very different from each other and who meet unexpectedly.

You should think about:

- Who are the characters and how are they different?
- Where, when and why do the characters meet each other?
- What do the characters do together, and what happens?

Space for your plan:

Write your story on the next page.

[25 marks]

